Experiment No. 2

QPSK Transmitter and Receiver using GNU Radio and PlutoSDR

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Aim

To design and implement a QPSK Transmitter and Receiver using GNU Radio and ADALM-PLUTO SDR for wireless digital communication.

Objective

The objective of this experiment is to study Quadrature Phase Shift Keying (QPSK) modulation and demodulation using Software Defined Radio. The experiment demonstrates digital data transmission, pulse shaping, carrier recovery, timing synchronization, and symbol decoding using GNU Radio and PlutoSDR.

Theory

Quadrature Phase Shift Keying (QPSK) is a digital modulation technique in which two bits are transmitted per symbol by changing the phase of the carrier signal. QPSK improves bandwidth efficiency compared to Binary Phase Shift Keying (BPSK).

In QPSK, four different phase states are used:

$$45^\circ,\ 135^\circ,\ 225^\circ,\ 315^\circ$$

Each phase represents a unique pair of bits.

The transmitted QPSK signal can be represented as:

$$s(t) = A\cos(2\pi f_c t + \phi)$$

Where:

Symbol	Meaning
A	Signal Amplitude
f_c	Carrier Frequency
ϕ	Phase Shift
t	Time

Table 1: Parameters used in QPSK Equation

In this experiment, random symbols are generated and mapped into QPSK constellation points. Root Raised Cosine (RRC) filters are used for pulse shaping and matched filtering. Symbol synchronization and carrier phase synchronization are achieved using Symbol Sync and Costas Loop blocks respectively.

Equipment Required

- ADALM-PLUTO SDR

- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

QPSK Transmitter Flowgraph

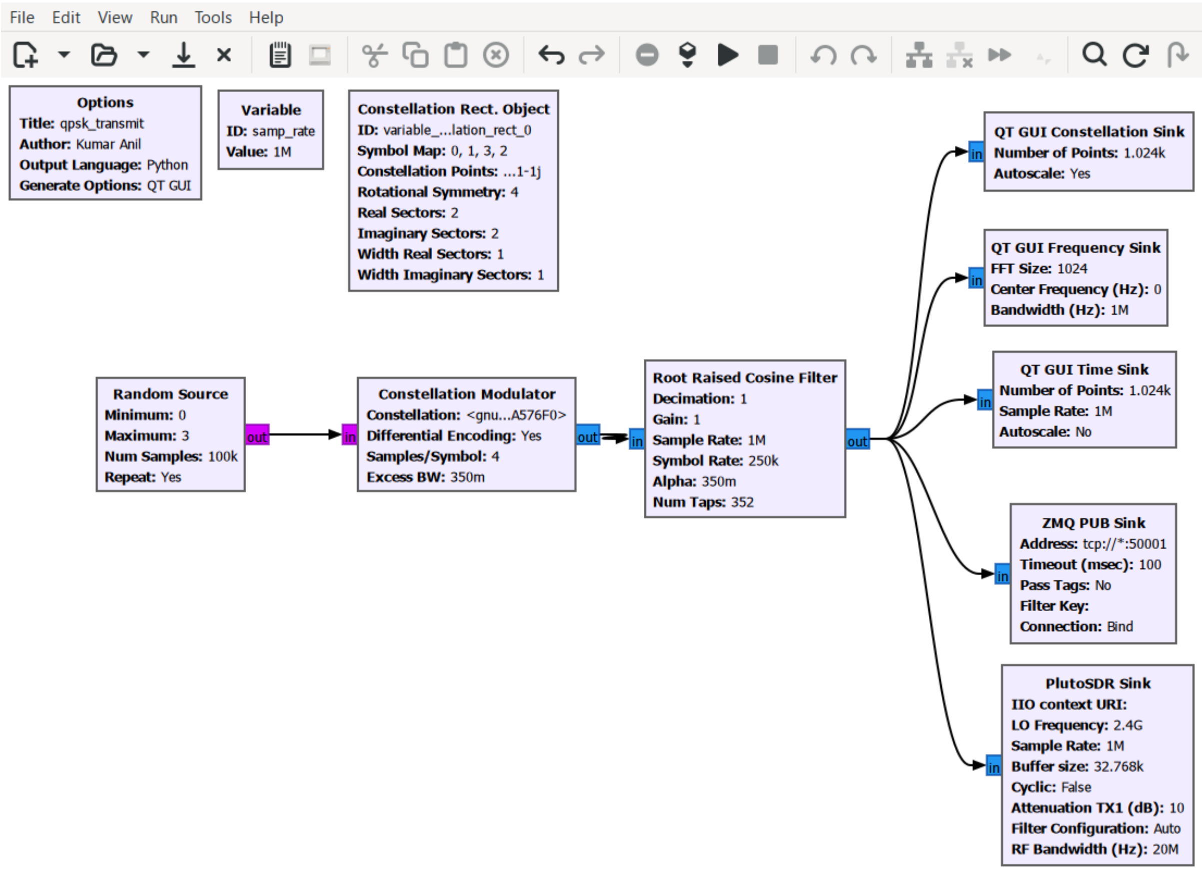

Figure 1: QPSK Transmitter Flowgraph

QPSK Receiver Flowgraph

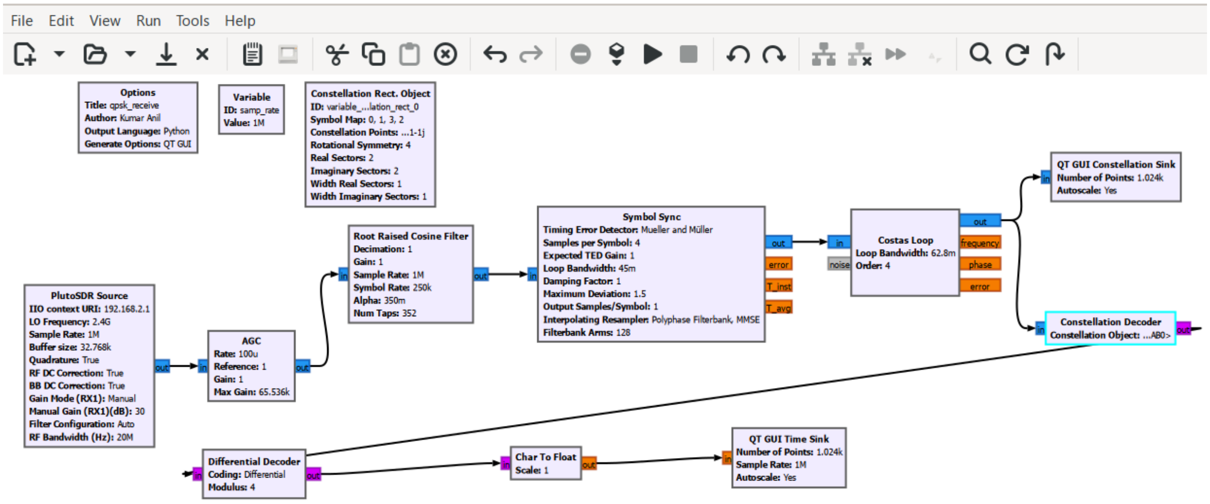

Figure 2: QPSK Receiver Flowgraph

Transmitter Description

The transmitter section begins with a Random Source block that generates random digital symbols. These symbols are mapped into QPSK constellation points using the Constellation Modulator block. Differential encoding is enabled to improve phase ambiguity handling. A Root Raised Cosine Filter performs pulse shaping and bandwidth limitation before transmission. The processed signal is visualized using QT GUI sinks and transmitted wirelessly using the PlutoSDR Sink at 2.4 GHz.

Receiver Description

The receiver section starts with the PlutoSDR Source which captures the transmitted RF signal. The AGC block stabilizes signal amplitude variations. A Root Raised Cosine Filter performs matched filtering and reduces intersymbol interference. Symbol synchronization is achieved using the Symbol Sync block with Mueller and Müller timing recovery. The Costas Loop block corrects carrier frequency and phase offsets. The corrected symbols are visualized using a Constellation Sink and decoded using the Constellation Decoder and Differential Decoder blocks. Finally, the recovered digital data is displayed using a QT GUI Time Sink.

Working Principle

The transmitter continuously generates random symbols and modulates them using QPSK modulation. The signal is pulse-shaped using an RRC filter and transmitted through PlutoSDR. At the receiver side, the incoming signal is filtered, synchronized, and phase-corrected. The received constellation points are decoded back into digital symbols using synchronization and decoding blocks. The developed QPSK communication architecture is hardware independent and can be adapted for interoperability between different SDR platforms such as PlutoSDR, USRP, and HackRF by maintaining compatible modulation, synchronization, and sampling parameters.

GNU Radio Blocks Used

Block	Function
Random Source	Generates random symbols
Constellation Modulator	Maps symbols to QPSK constellation
RRC Filter	Pulse shaping and matched filtering
PlutoSDR Sink	RF transmission
PlutoSDR Source	RF reception
AGC	Amplitude normalization
Symbol Sync	Timing recovery
Costas Loop	Carrier phase synchronization
Constellation Decoder	Symbol recovery
QT GUI Sinks	Signal visualization

Table 2: GNU Radio Blocks Used

Parameters

Parameter	Value
Sample Rate	1 MS/s
Carrier Frequency	2.4 GHz
Samples per Symbol	4
Excess Bandwidth	0.35
RRC Filter Taps	352
Constellation Order	4
Differential Encoding	Enabled
RX Gain Mode	Manual

Table 3: QPSK System Parameters

Procedure

1. Open GNU Radio Companion.
2. Create the QPSK transmitter flowgraph using Random Source, Constellation Modulator, Root Raised Cosine Filter, and PlutoSDR Sink.
3. Configure the sample rate as 1 MS/s and carrier frequency as 2.4 GHz.
4. Enable differential encoding in the Constellation Modulator block.
5. Add QT GUI Frequency Sink and Constellation Sink to monitor transmitted signals.
6. Create the receiver flowgraph using PlutoSDR Source, AGC, RRC Filter, Symbol Sync, Costas Loop, and Constellation Decoder.
7. Configure Symbol Sync for Mueller and Müller timing recovery.
8. Configure the Costas Loop for carrier phase synchronization.
9. Connect the transmitter and receiver using PlutoSDR hardware.
10. Run both flowgraphs.
11. Observe the constellation diagram and verify stable constellation points.
12. Analyze the decoded output signal in the Time Sink.

Observation

Stable constellation points were observed after carrier and timing synchronization. The AGC block normalized the received signal amplitude, while the Symbol Sync and Costas Loop corrected symbol timing and carrier phase errors. The decoded data matched the transmitted symbol stream.

Applications

QPSK modulation is widely used in wireless communication systems, satellite communication, Wi-Fi systems, digital television broadcasting, SDR systems, and modern digital communication networks.

Conclusion

The QPSK Transmitter and Receiver were successfully implemented using GNU Radio and PlutoSDR. The experiment demonstrated digital modulation, synchronization, carrier recovery, and wireless communication using SDR technology.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. Digital Communication Systems
4. SDR and Wireless Communication Principles